

METU Computer Engineering

CEng 111 – Fall 2026
Week 4b

Credit: Some slides are from the “Invitation to Computer Science” book by G. M. Schneider, J. L. Gersting and some from the “Digital Design” book by M. M. Mano and M. D. Ciletti.

Previously on CENG111!

Memory Types

- Dynamic Memory
  - The voltages stored in capacitors die away with time.
  - Solution?
    - Refresh the memory frequently; i.e., re-write the contents of the memory.
- Static Memory
  - A coupled transistor system stores the information.
  - One of the couples *triggers* the other.
- DRAM is cheaper and more widely used.

http://wiki.xtronics.com/index.php/How_Memory_Works



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Memory Types

- Volatile Memory:
  - The stored values are lost when power is off.
  - DRAM, SRAM
- Non-volatile:
  - Read-only Memory (ROM), flash memory

http://wiki.xtronics.com/index.php/How_Memory_Works



Previously on CENG111!

PERIPHERALS

* You are not responsible from this in the exam.



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INTERRUPTS

* You are not responsible from this in the exam.



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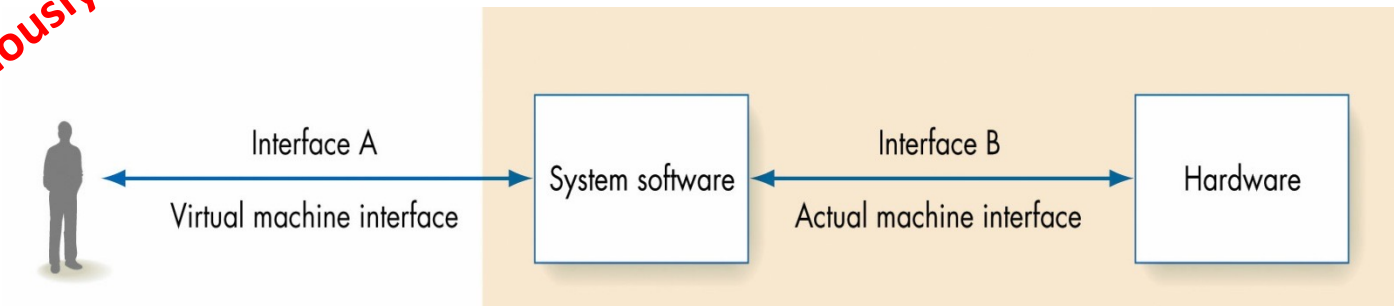
What happens after that power button

BOOTING YOUR COMPUTER

* You are not responsible from this in the exam.

For more details, see e.g.: https://neosmart.net/wiki/mbr-boot-process/

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OPERATING SYSTEM

* You are not responsible from this in the exam.

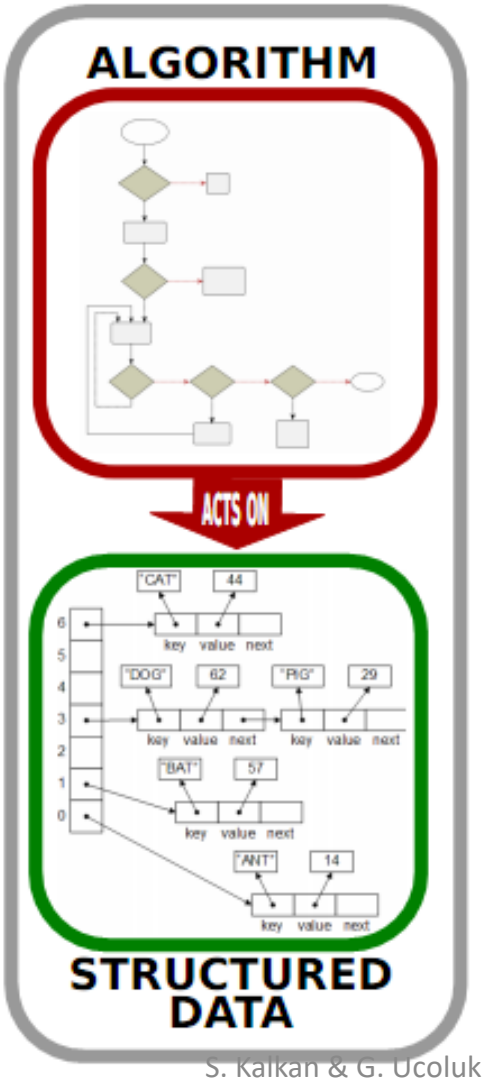
Program, Programming

Previously on CENG111!

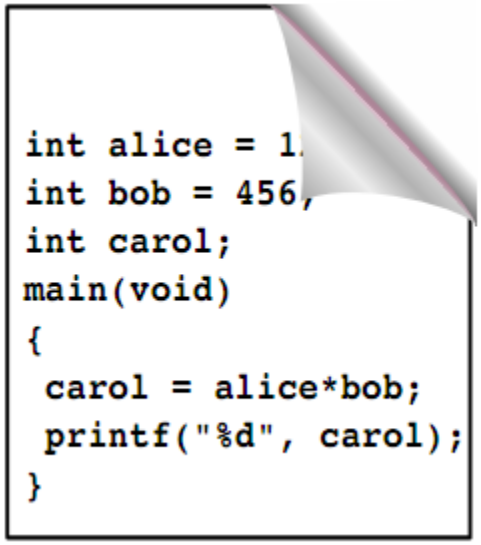


WORLD PROBLEM

TRANSFORMED



IMPLEMENTED

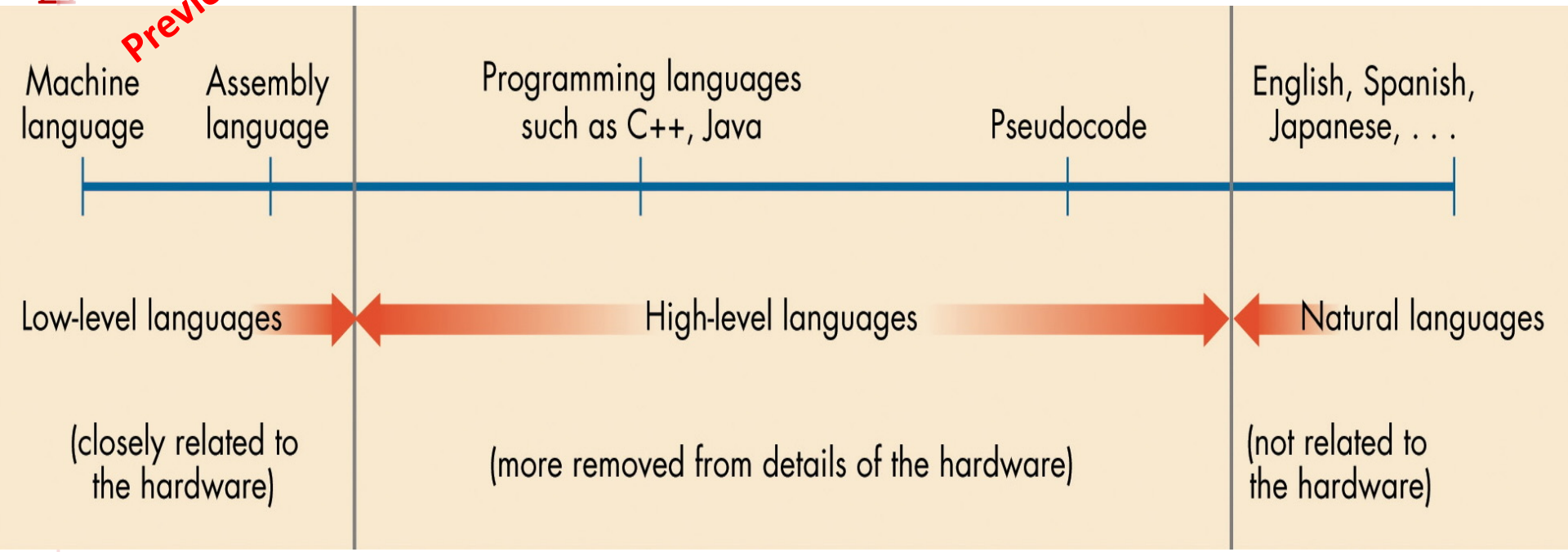


```
int alice = 1
int bob = 456,
int carol;
main(void)
{
 carol = alice*bob;
 printf("%d", carol);
}
```

PROGRAM

The Spectrum of Programming Languages

Previously on CENG111!



- There is a limit to how high a language can get.
- Why can't we write programs in our spoken language?



Today

- Continue with the World of Programming
  - Zoo of Programming Languages (PLs)
  - Choosing a PL
  - How PLs are implemented
  - Testing & Programming errors
  - Meet Python

Administrative Notes

- Von Neumann Machine Exam:
  - Preliminary Text has been announced!
  - In-class Exam: 23 October (Friday) – 18:00



Zoo of Programming Languages

- Around 700 programming languages!

- But, why do we not have a programming language that serves all paradigms/purposes/requirements?

Choosing a PL

- Ex: Moving soil with a shovel and a grader







Factors that affect choosing a PL

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- Domain & Technical Nature of the Problem

a) Finding the pixels of an image with RGB value of [202,130,180] with a tolerance of 5.4% in intensity.

b) A proof system for planar geometry problems.

c) A computer game platform which will be used as a whole or in parts and may get extended even rewritten by programmers at various levels.

d) Payroll printing.

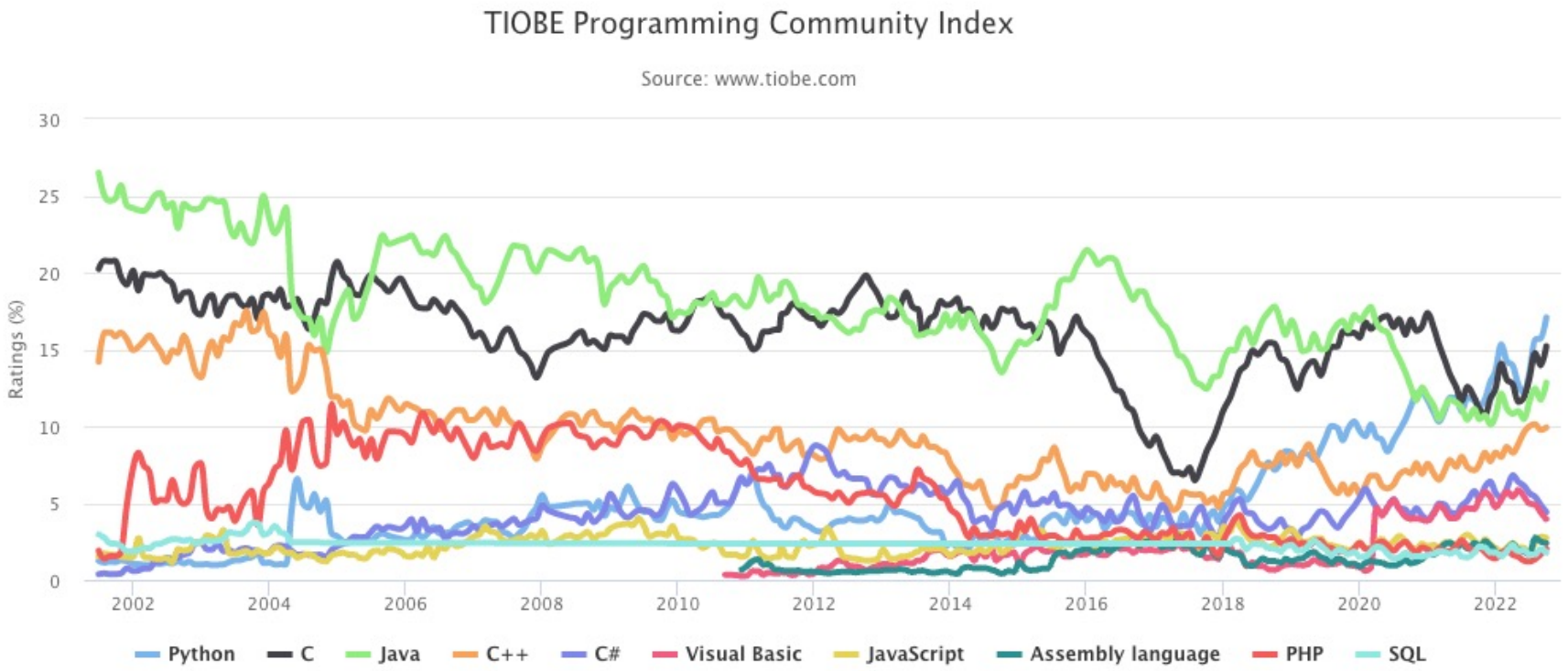


Factors that affect choosing a PL

- Personal taste and preference
- Circumstance-imposed constraints
  - e.g., time limit.
- Current trend

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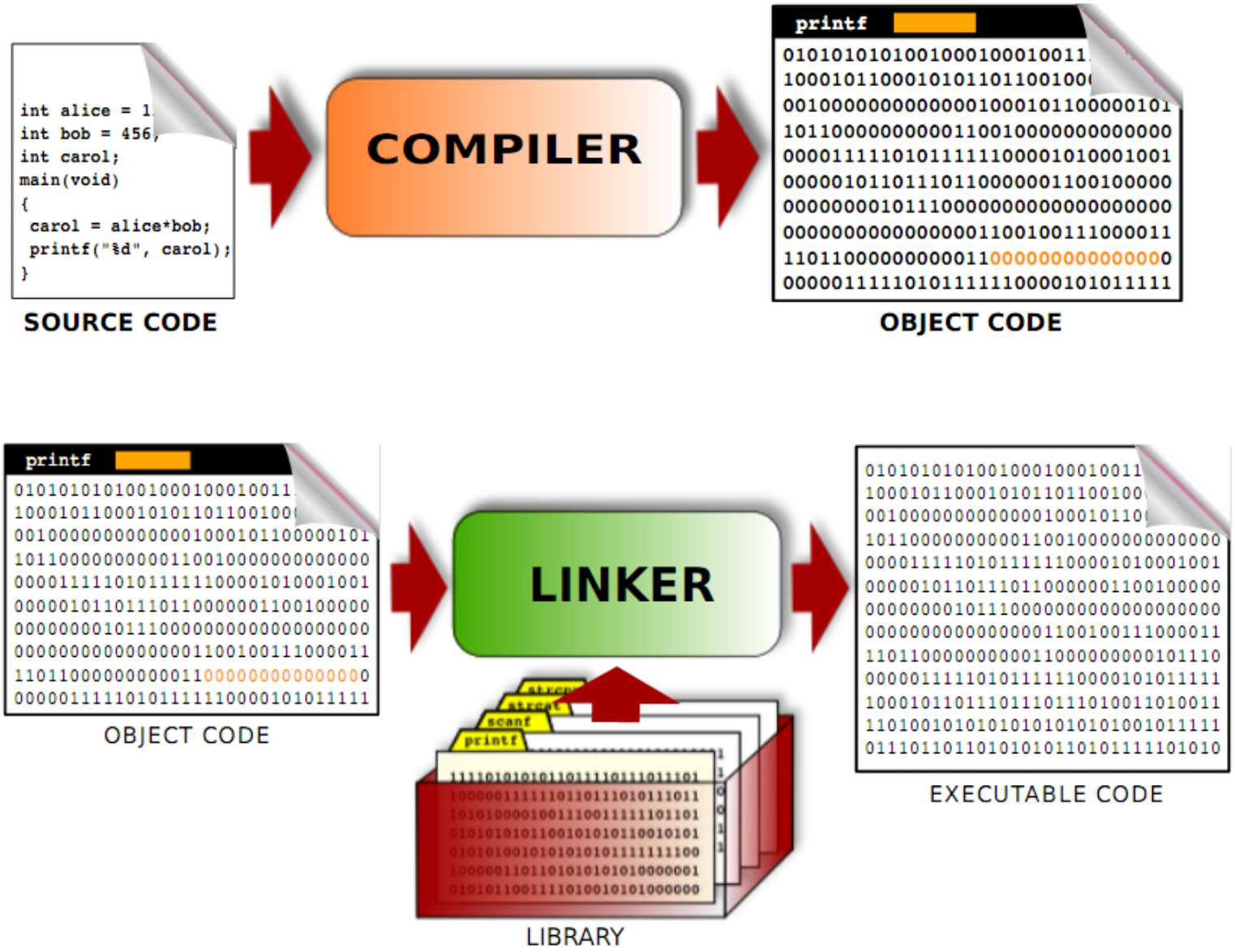
Current trend



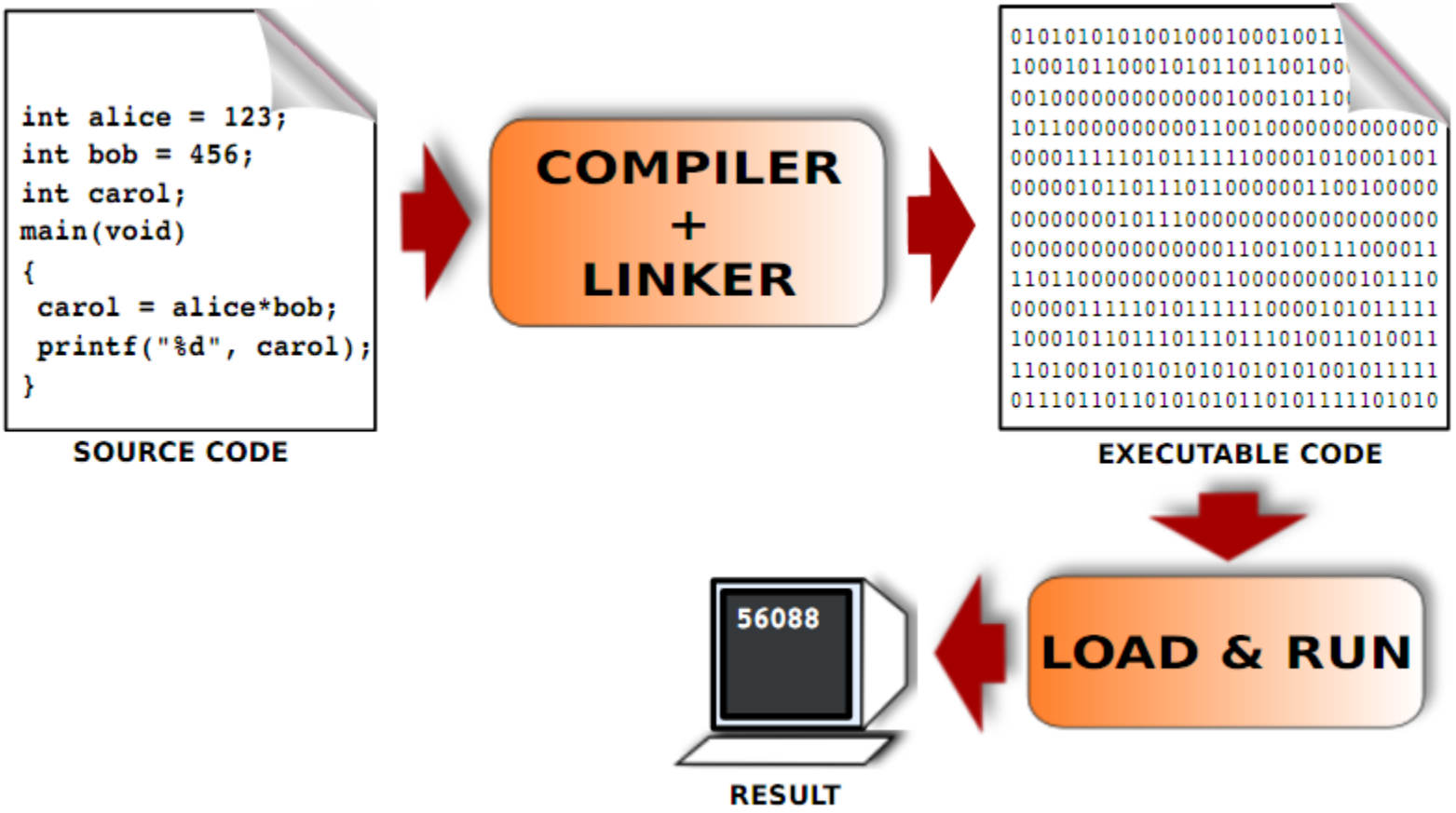
http://www.tiobe.com/index.php/content/paperinfo/tpci/index.html

How are languages implemented

COMPILATIVE APPROACH

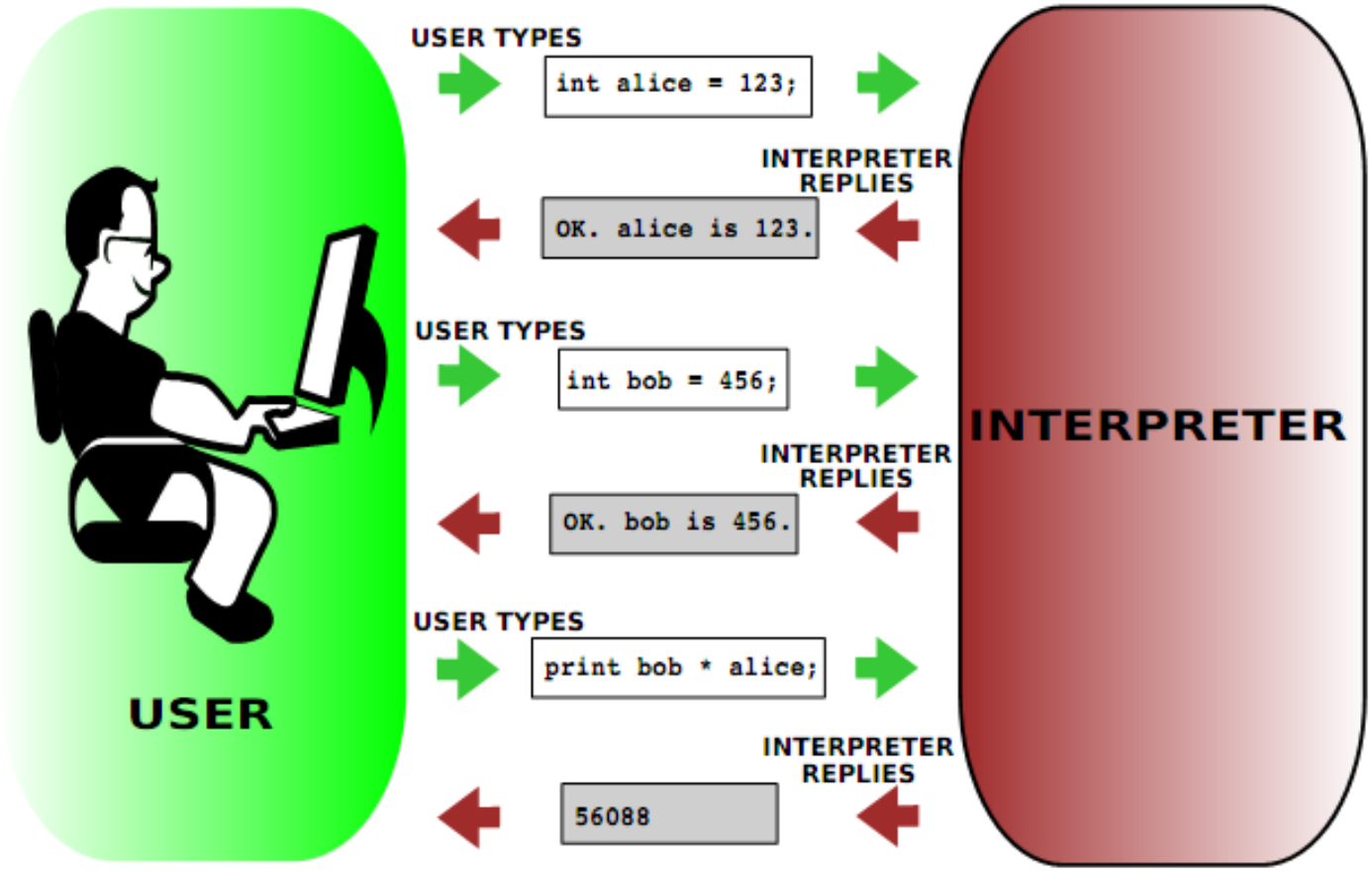


How are languages implemented



How are languages implemented

INTERPRETIVE APPROACH



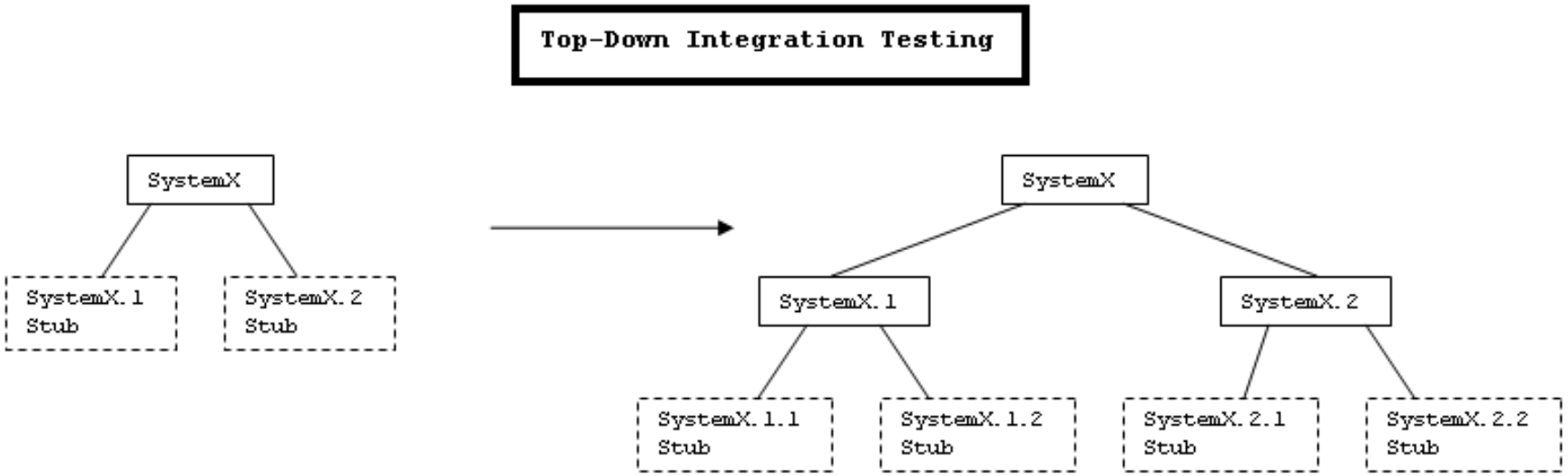


How is a program written?

- Modular & Functional Breakdown
- For example:
  - User interface module
  - Database module
  - Control module

Testing

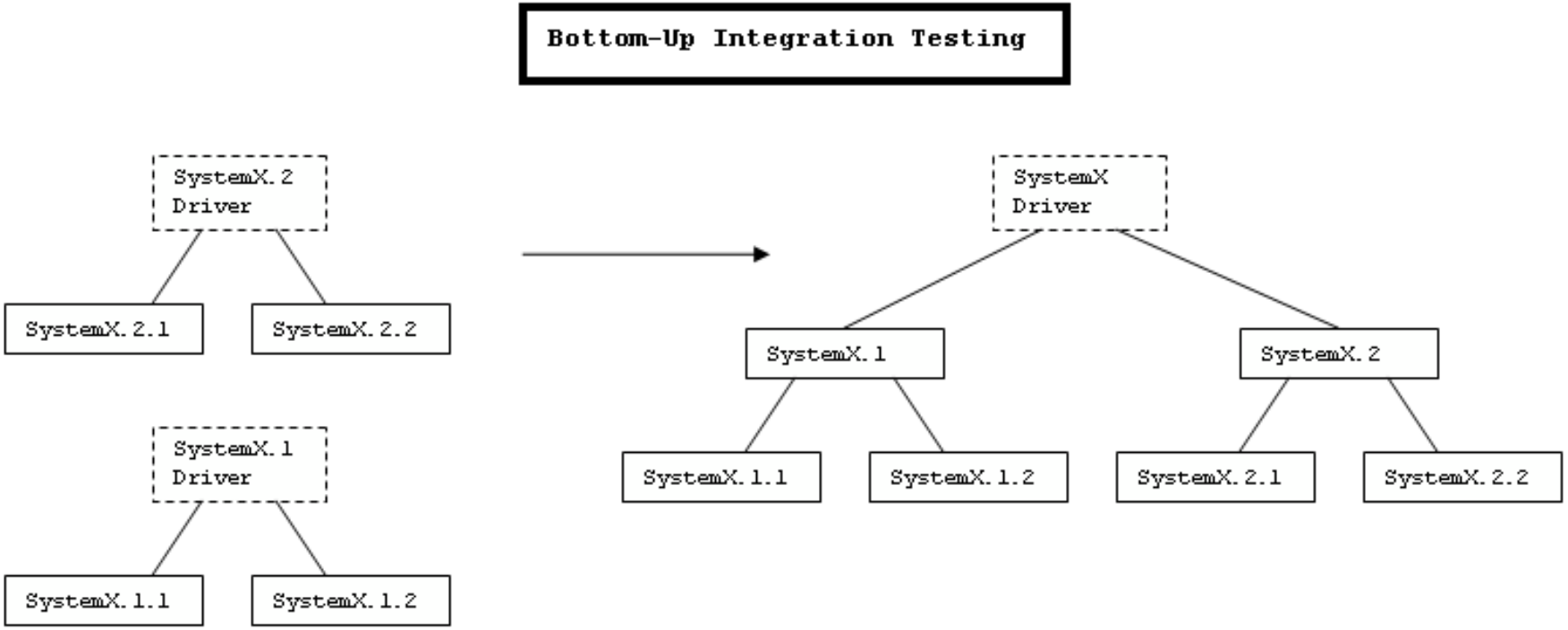
- Top-down Testing



http://sce.uhcl.edu/whiteta/sdp/subSystemIntegrationTesting.html

Testing

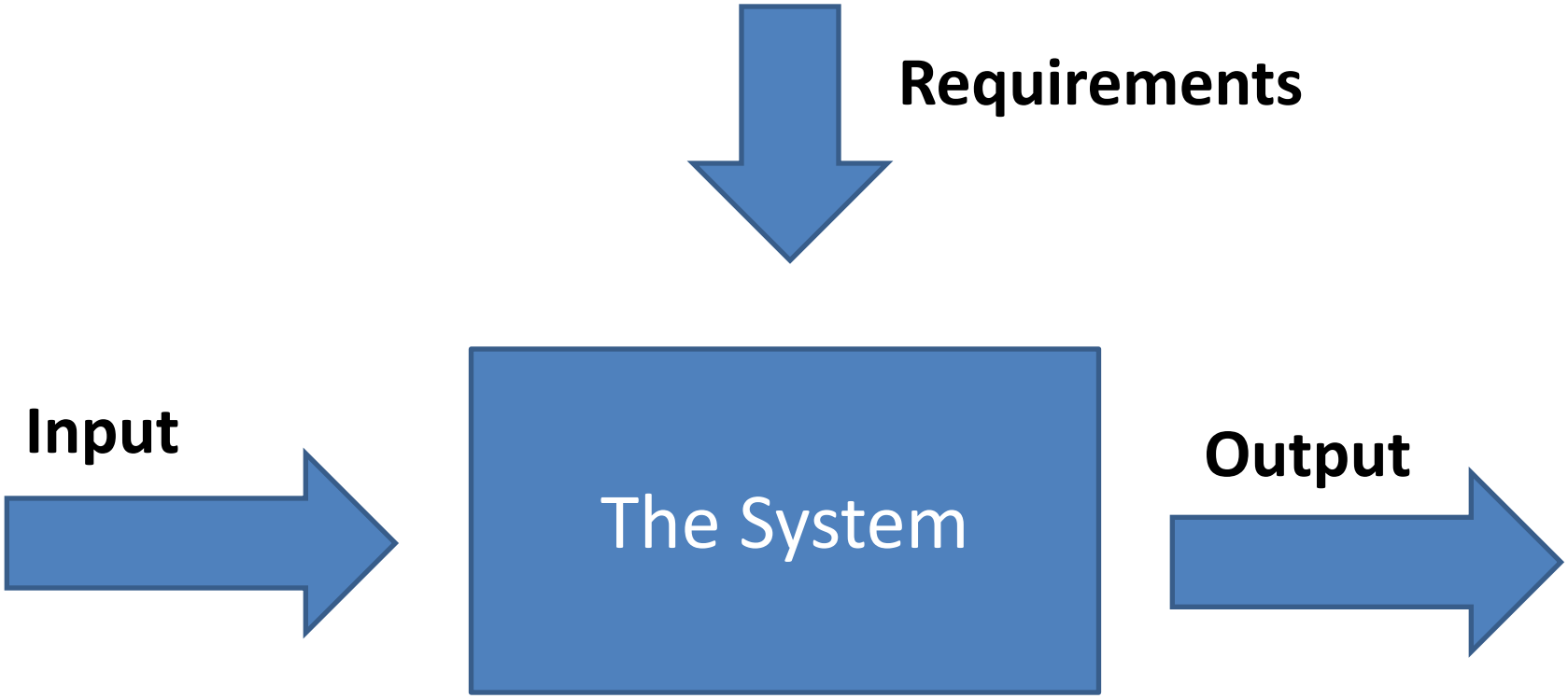
- Bottom-up Testing



http://sce.uhcl.edu/whiteta/sdp/subSystemIntegrationTesting.html

Testing

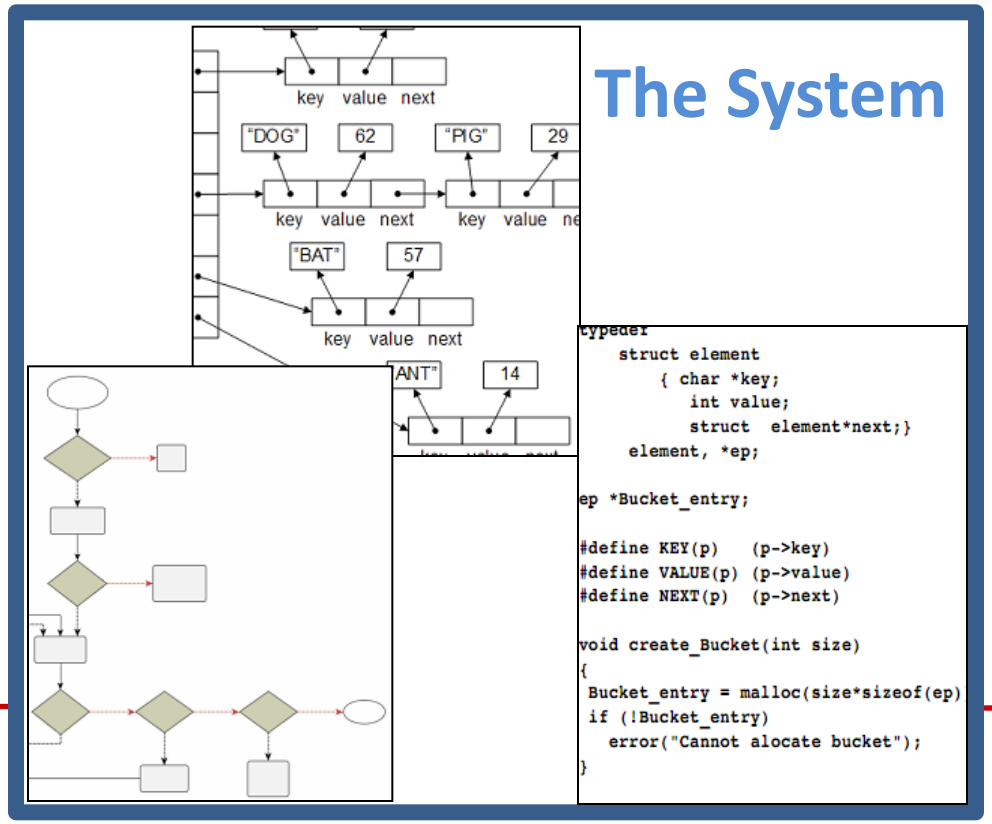
- Black-box Testing



Testing

- White-box Testing

Requirements



Input

Output

Bugs, Errors

- Syntax Errors

  Area = 3.1415 * R * R

  Area = 3.1415 x R x R

- Run-time Errors

```
>>> def SqrtDelta(a,b,c):
>>>       return sqrt(b*b - 4*a*c)
>>>
>>> print SqrtDelta(1,3,1)
2.236067977497898
>>> print SqrtDelta(1,1,1)
ValueError: math domain error
```

Bugs, Errors

- Logical Errors

$$root_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$



```
>>> root1 = (- b + sqrt(b*b - 4*a*c)) / 2*a
```

- Design Errors

$$x^3 + ax^2 + bx + c = 0$$

$$root_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

MEET python™



Guido van Rossum (1956 -)



- An interpretive/scripting PL that:
  - Longs for code readability
  - Ease of use, clear syntax
  - Wide range of applications, libraries, tools
- Zen of Python [https://en.wikipedia.org/wiki/Zen_of_Python]
  - Beautiful is better than ugly.
  - Explicit is better than implicit.
  - Simple is better than complex.
  - Complex is better than complicated.
  - Flat is better than nested.
  - Sparse is better than dense.
  - Readability counts.
  - …



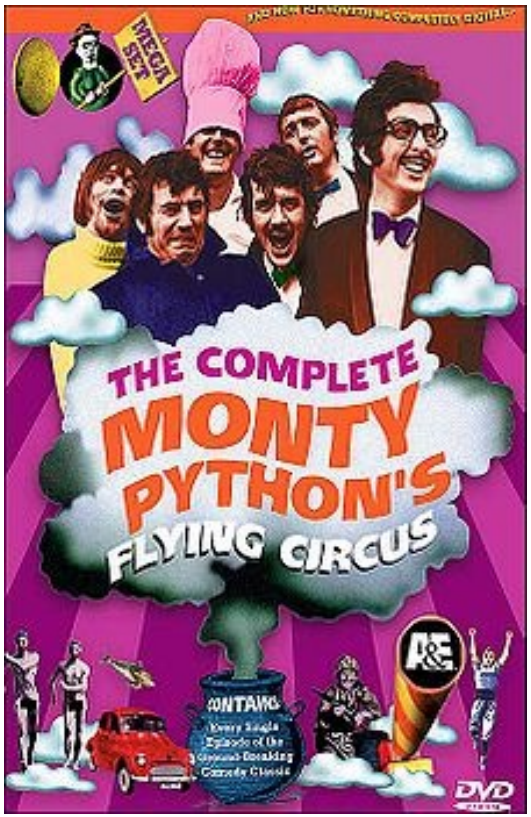
- Supports multiple paradigms:
  - Imperative
  - Functional
  - Object-oriented



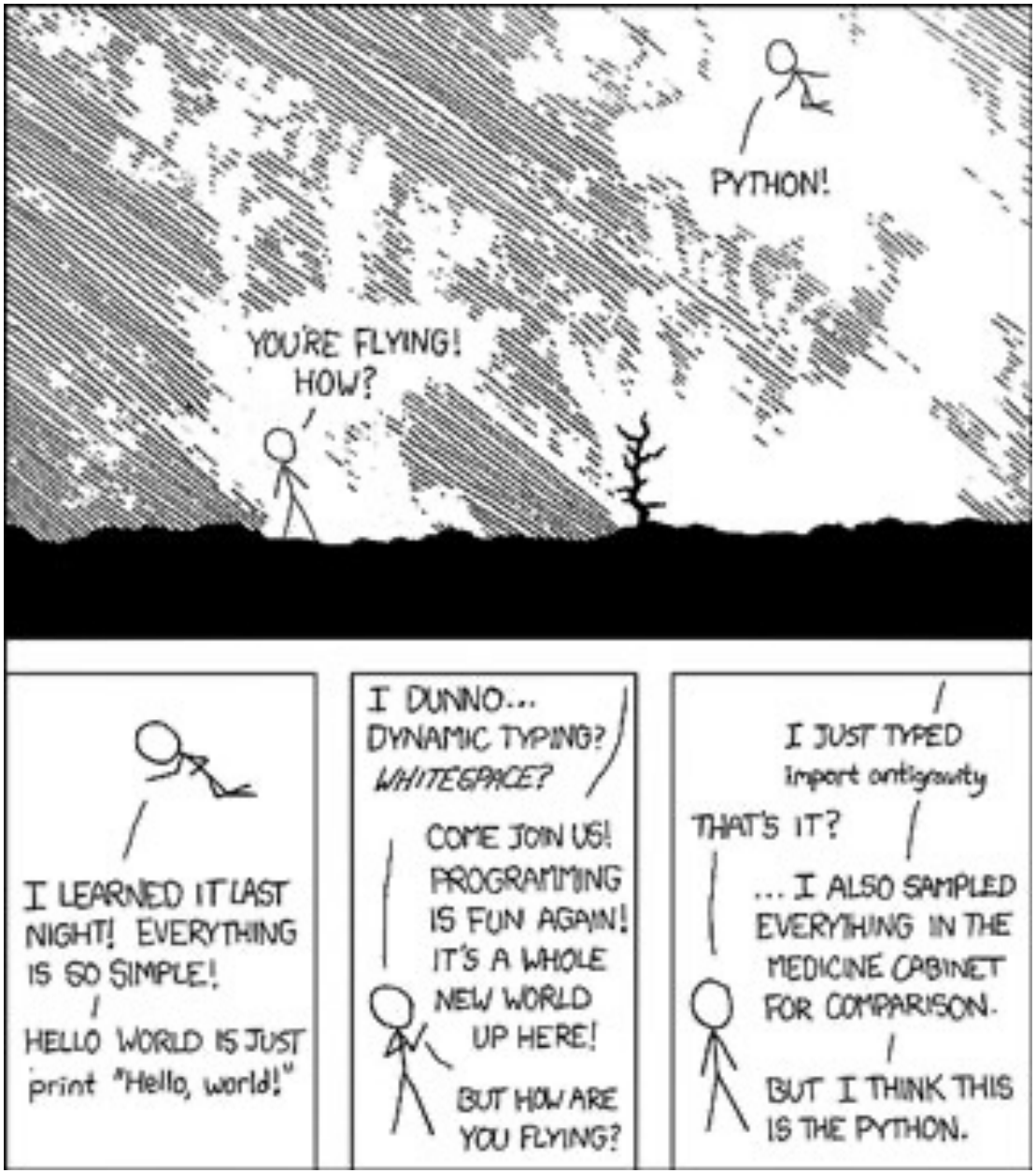
- Started at the end of 1980s.
- V2.0 was released in 2000
  - With a big change in development perspective: Community-based
  - Major changes in the facilities.
- V3.0 was released in 2008
  - Backward-incompatible
  - Some of its features are put into v2.6 and v2.7.

python™

- Where does the name come from?
  - While van Possum was developing Python, he read the scripts of Monty Python's Flying Circus and thought 'python' was "short, unique and mysterious" for the new language [1]
- One goal of Python: "fun to use"
  - The origin of the name is the comedy group "Monty Python"
  - This is reflected in sample codes that are written in Python by the original developers.



[1] https://docs.python.org/2/faq/general.html#why-is-it-called-python


PYTHON!
YOU'RE FLYING! HOW?
I LEARNED IT LAST NIGHT! EVERYTHING IS SO SIMPLE!
HELLO WORLD IS JUST print "Hello, world!"
I DUNNO... DYNAMIC TYPING? WHITESPACE?
COME JOIN US! PROGRAMMING IS FUN AGAIN! IT'S A WHOLE NEW WORLD UP HERE!
BUT HOW ARE YOU FLYING?
I JUST TYPED import antigravity
THAT'S IT?
... I ALSO SAMPLED EVERYTHING IN THE MEDICINE CABINET FOR COMPARISON.
BUT I THINK THIS IS THE PYTHON.




```
skalkan@divan:~$ python
Python 2.5.2 (r252:60911, Jan 24 2010, 17:44:40)
[GCC 4.3.2] on linux2
Type "help", "copyright", "credits" or "license" for more information.
>>>
```